

ORIGINAL ARTICLE

Integrative Proteomic Profiling of Blood Pressure and Hypertension

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BACKGROUND: Hypertension affects one-third of adults worldwide and is largely idiopathic. Identifying protein biomarkers of hypertension may reveal underlying mechanisms and potential treatments.

METHODS: We examined associations between 2922 plasma proteins and blood pressure (BP) traits (systolic BP [SBP] and diastolic BP [DBP], and prevalent hypertension) in 45991 UK Biobank participants with validation in 5759 FHS (Framingham Heart Study) participants. Analysis of new-onset hypertension was performed in 3995 UK Biobank participants and validated in 3073 FHS participants. Functional enrichment analysis linked proteins to biological functions and diseases. Mendelian randomization assessed causal relations between proteins using *cis*-protein quantitative trait loci as exposures and BP traits as outcomes.

RESULTS: In the UK Biobank, 342 proteins were associated (false discovery rate knockoff <0.1) with SBP, 276 with DBP, and 200 with prevalent hypertension ($N=24\,724$). In the FHS, 115 proteins were validated (Bonferroni-corrected $P<0.05$) for SBP, 108 for DBP, and 102 for hypertension. New-onset hypertension was associated with 60 (17 validated) proteins. Mendelian randomization analyses revealed significant associations for 206, 210, and 82 proteins with SBP, DBP, and hypertension, respectively. Several of these proteins are involved in blood vessel and cardiac muscle morphogenesis, vasoconstriction, and inflammation. BP-related proteins are enriched for atherosclerotic cardiovascular, liver, and kidney diseases. NHERF2 (Na^+/H^+ exchange regulatory cofactor 2) demonstrated putatively causal associations with SBP, DBP, and with prevalent and new-onset hypertension.

CONCLUSIONS: This large-scale proteomic study revealed protein signatures of BP traits and hypertension risk, highlighting inflammation and cardiac and vascular remodeling processes; NHERF2 and several other proteins emerged as promising therapeutic targets. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.26407.)

• **Supplement Material.**

Key Words: biomarkers ■ blood pressure ■ hypertension ■ Mendelian randomization analysis ■ proteomics

Hypertension represents a key modifiable risk factor for cardiovascular disease, kidney disease, and premature death.¹ Over 1.3 billion adults worldwide have hypertension, with nearly half of them unaware of their condition.² With fewer than 25% of all individuals with hypertension successfully managing to control their condition,^{2,3} there is an urgent need for enhanced preventive and treatment strategies to alleviate the burden of hypertension-related health outcomes. Although recent declines in hypertension control are largely attributed to gaps in implementation and health system delivery,^{4,5} continued efforts to refine prevention and treatment

approaches remain important. It is crucial to identify molecular mechanisms contributing to the development of hypertension as a prelude to the development of more effective prevention and treatment strategies.

Identifying circulating protein signatures of blood pressure (BP) and hypertension via high-throughput proteomics could reveal a diverse panel of novel biomarkers involved in BP regulation and the physiological response to hypertension. These biomarkers may provide insights into the mechanisms involved in the pathophysiology of hypertension and hypertension-related diseases; those that are causally linked may serve as promising

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NOVELTY AND RELEVANCE

What Is New?

We screened 2922 circulating proteins to uncover biomarkers associated with systolic and diastolic blood pressure (BP), as well as preexisting and new-onset hypertension, using a robust method to control for false discovery rate. Previous BP protein-association studies investigated far fewer candidate proteins, lacked external validation of findings, or failed to use rigorous statistical measures to address inflated results due to high intercorrelation among protein predictors. By integrating these findings with recent large genome-wide association study data sets, we identified 15 plasma proteins with evidence for a putatively causal role in BP regulation and hypertension. This study also highlighted disease pathways and comorbidities associated with hypertension through enrichment of protein signatures.

What Is Relevant?

Hypertension, despite being a key modifiable risk factor for cardiovascular disease, is poorly controlled in 3 quarters of affected individuals. Protein biomarkers discovered through integrative genomic approaches that display a putative causal role in BP regulation are promising therapeutic candidates. Identification of key biological pathways and comorbidities associated with hypertension can enhance our understanding of the disease and approaches to management.

Clinical/Pathophysiological Implications?

This study identifies plasma proteins with putative casual association with BP and hypertension, thereby providing a prioritized list of candidate proteins that warrant further investigation as therapeutic targets for BP control and hypertension management.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
APOA1	apolipoprotein A-I
APOE	apolipoprotein E
BMI	body mass index
BP	blood pressure
CHMP1A	charged multivesicular body protein 1A
cis-pQTL	cis-protein quantitative trait locus
CTH	cystathionine gamma-lyase
DBP	diastolic blood pressure
EDH4	EH domain-containing protein 4
FDR	false discovery rate
FGF5	fibroblast growth factor 5
FHS	Framingham Heart Study
FN1	fibronectin
GWAS	genome-wide association study
IGFBP3	insulin-like growth factor-binding protein 3
LRIG1	leucine-rich repeats and immunoglobulin-like domains protein 1
LTBP2	latent-transforming growth factor β -binding protein 2
MAPK	mitogen-activated protein kinase
MR	Mendelian randomization
NADK	NAD kinase
NECAB2	N-terminal EF-hand calcium-binding protein 2
NHERF2	Na ⁺ /H ⁺ exchange regulatory cofactor 2
PLA2G1B	phospholipase A2

SBP	systolic blood pressure
TIE1	tyrosine-protein kinase receptor Tie-1
UKB	UK Biobank
VTN	vitronectin
WARS1	tryptophan-tRNA ligase, cytoplasmic

therapeutic targets. Previous protein-association studies of hypertension were constrained by their reliance on a limited set of proteins that may have been insufficient to study the breadth of pathways implicated in hypertension.^{6–8} Hypertension is characterized by an array of physiological perturbations,⁹ including endothelial dysfunction, aberrant renal sodium regulation, systemic inflammation, elevated oxidative stress, reduced fibrinolytic potential, vascular hypertrophy, and dysregulation of neurohormonal and adrenergic pathways. The use of high-throughput proteomics and bioinformatic techniques could reveal additional pathways involved in hypertension, and disease-risks associated with it. A recent proteomics study investigated biomarkers associated with BP and hypertension;¹⁰ however, it relied on older genetic data sets for causal inference analysis, and the association results showed highly inflated *P* values. It also lacked external validation of biomarker findings, which is a critical step in omics research.¹¹

To this end, we analyzed 2922 plasma proteins to uncover associations with systolic BP (SBP), diastolic BP (DBP), and prevalent and new-onset hypertension in UK Biobank (UKB) participants (Figure 1). To control for markedly inflated *P* values, we applied a knock-off filter to the association analyses to control the false

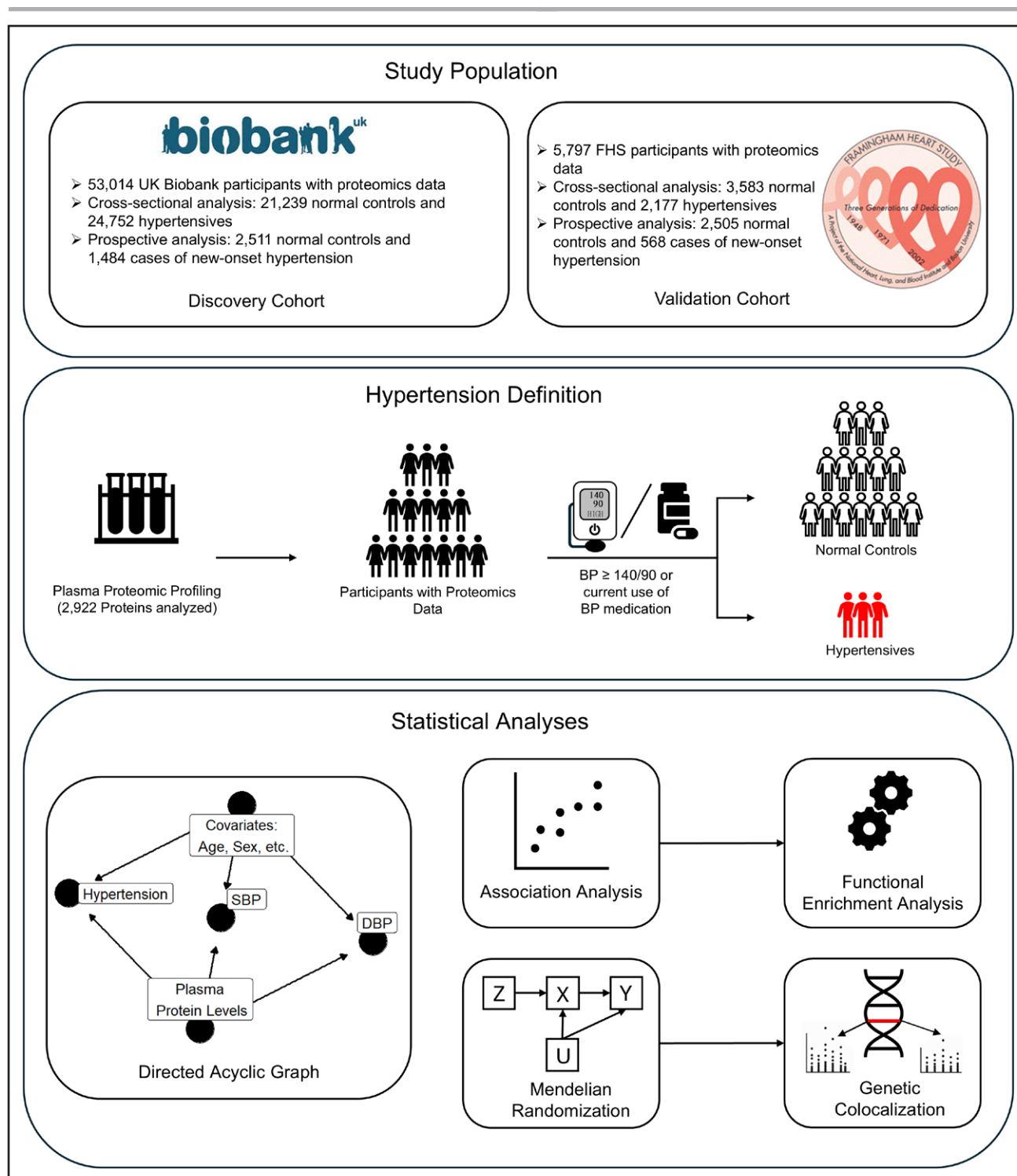


Figure 1. Conceptual framework of the study.

BP indicates blood pressure; DBP, diastolic blood pressure; FHS, Framingham Heart Study; and SBP, systolic blood pressure.

discovery rate (FDR).¹² We validated discovery associations in an independent external study sample from the FHS (Framingham Heart Study), ensuring robustness of our findings. In secondary analyses, we tested 2538 additional proteins that were available only in FHS for association with BP traits. In addition, we

used Mendelian randomization (MR) and genetic colocalization analyses to identify proteins with putatively causal relations and shared genetic signals with BP traits. By unraveling proteomic signatures of BP and hypertension, this study uncovered numerous novel BP biomarkers and provides mechanistic insights into

the pathophysiology of hypertension. The identification of causal protein biomarkers could be instrumental in the development of new and effective antihypertensive interventions.

METHODS

Data Availability

FHS data are available on the database of Genotypes and Phenotypes, accession no. phs000007 (https://www.ncbi.nlm.nih.gov/gap/advanced_search/?TERM=phs000007).

UKB proteomics and phenotype data are available through the UK Biobank Research Analysis Portal (<https://www.ukbiobank.ac.uk/enable-your-research>).

Genome-wide association study (GWAS) summary data used for MR and colocalization analyses for SBP and DBP based on Keaton et al¹³ are publicly available on the GWAS Catalog website data repository (<https://www.ebi.ac.uk/gwas>). GWAS summary data for *cis*-pQTL (*cis*-protein quantitative trait locus) results and hypertension are available online through an interactive portal (<http://ukb-ppp.gwas.eu>) and the MRC IEU repository (<https://gwas.mrcieu.ac.uk>), respectively.

Study Sample

For the cross-sectional analyses, the discovery set included 45 991 UKB participants who underwent assessment using the Olink Explore 3072 proteomic assay (Olink, Inc, Uppsala, Sweden) as part of the UKB-Pharma Proteomics Project.¹⁴ The validation set included 5760 FHS participants with proteomic data using the Olink Explore HT panel¹⁵ (Figure 1). Individuals who were missing BP readings or covariate data were excluded from the analyses. For new-onset hypertension analysis in UKB, we additionally excluded 24 752 participants with hypertension at baseline and 17 244 without follow-up BP data, yielding a final analytic sample of 3995 individuals. These results were validated in 3073 FHS participants. The UKB protocol was approved by the North West Multi-Center Research Ethics Committee. The FHS study protocol was approved by the Boston University Medical Campus institutional review board. All included participants provided written informed consent.

BP Measurement and Definition of Prevalent and Incident Hypertension

In UKB, 2 automated measures of SBP and DBP were taken with a 1-minute interval using an Omron 705 IT electronic BP monitor (Omron Healthcare, Inc, Kyoto, Japan) at baseline (2006–2010) and during repeat assessments (time-to-event; median=10.1 years, interquartile range=5.4 years), after 5 minutes of seated rest. In FHS, BP measurements were recorded twice by the examining physician using a standardized protocol and a mercury column sphygmomanometer at the baseline and follow-up examinations (time-to-event; median=8 years, interquartile range=1 year). The mean of 2 BP readings obtained 5 minutes apart was used for the analysis. We considered individuals hypertensive if they had either elevated BP (SBP $\geq$ 140 mm Hg or DBP $\geq$ 90 mm Hg) or were currently treated with antihypertensive medication. For prospective analysis, we defined incident hypertension as the occurrence of hypertension at the

follow-up examinations in a participant who was normotensive at baseline.

Protein Quantification

Plasma levels of 2923 proteins were quantified as normalized protein expression using the Olink Explore 3072¹⁶ panel in UKB and 5367 proteins using the Olink Explore HT¹⁵ panel in FHS participants. Both Olink Explore 3072 and Explore HT are based on proximity extension assays; over 95% of assays in Explore 3072 are also included in Explore HT, and the 2 platforms yield highly comparable results. The Olink proteomics assay protocol in UKB is described in detail elsewhere.¹⁴ The association analysis of plasma proteins and BP traits was conducted for 2922 proteins (1 protein was excluded due to excessive missingness) in the UKB discovery set. In the FHS, 5343 proteins were analyzed, with 2805 proteins overlapping with those measured in UKB (Supplemental Material). The remaining 2538 proteins in FHS participants were unavailable in UKB and were analyzed separately.

Statistical Analysis

R, version 4.5.0 was used for data analysis and visualization. We examined cross-sectional associations of plasma proteins with continuous (SBP and DBP) and prevalent hypertension using linear and logistic regression, respectively. Proportional hazards regression was used to identify proteomic associations with new-onset hypertension. In the primary analysis, we adjusted for the fixed effects of covariates including age, age², sex, age $\times$ sex interaction, smoking status (current/former/never), alcohol consumption (ounces of ethanol per month) and estimated glomerular filtration rate. Estimated glomerular filtration rate was calculated in UKB using the 2021 CKD-EPI race-independent creatinine–cystatin C equation,¹⁷ and in the FHS using the 2009 CKD-EPI creatinine-based equation.¹⁸ The residuals of protein expression, obtained by regressing out the batch effects (with relevant batch variables treated as random variables) and familial relatedness (for FHS only), were utilized as the predictors. A threshold of $P < 0.000017$, reflecting Bonferroni correction (0.05/2922) for testing of 2922 proteins, was used to define statistical significance in the UKB discovery analysis. Under this conservative threshold, our sample size provided adequate statistical power ($\geq 80\%$) to detect even very small effect sizes, with near-complete power for small-to-large effects (Supplemental Material). We assessed the inflation of P values in the UKB primary analysis by calculating the genomic inflation factor (λ_{GC}) and visually inspecting the quantile-quantile plots. Owing to pronounced P value inflation observed in cross-sectional analysis (Figure S1), likely driven by strong intercorrelations among protein predictors, we applied the reflection knockoff filter,¹² combined with elastic-net regression, to analyze all proteomic features jointly and to control the FDR more rigorously. Knockoff-estimated FDR_{knockoff} < 0.1 defined statistical significance in discovery analysis for SBP, DBP, and prevalent hypertension. For external validation in FHS, significance was set at $P < 0.05$ divided by the number of proteins that were significant in UKB for a given trait and available for testing in FHS (using linear, logistic regression, or proportional hazards regression, as appropriate). In addition, we also identified proteins that were nominally significant in the

FHS, but these were not considered validated. In an exploratory analysis of additional 2538 proteins measured only in FHS, significance was defined as $P < 9.4 \times 10^{-6}$, reflecting Bonferroni correction for all 5343 proteins measured in FHS.

For sensitivity analyses, we applied a standard medication-adjustment approach by adding 15 mmHg to SBP and 10 mmHg to DBP among participants receiving antihypertensive therapy,¹⁹ to account for treatment-related lowering of measured BP. This approach addresses potential bias in BP estimation due to medication use. Second, when analyzing hypertension as a binary outcome, we repeated the analysis after redefining hypertension as SBP/DBP $\geq 130/80$ mmHg or current use of antihypertensive drugs, and we additionally conducted analyses excluding individuals taking antihypertensive drugs. Finally, since we did not adjust for body mass index (BMI) in the primary model because it is a strong causal risk factor for hypertension,²⁰ we included BMI as an additional covariate and reanalyzed SBP, DBP, and both prevalent and incident hypertension to reveal proteins associated with BP independent of adiposity.

We performed enrichment analysis to identify the overrepresented gene sets corresponding to the proteins significant in the primary analysis in the UKB. The top overrepresented gene sets, assessed using hypergeometric tests with significance set at Bonferroni-adjusted $P < 0.05$, were used to infer the biological roles of the proteins associated with BP traits. We utilized the Gene Ontology database²¹ (clusterProfiler package in R) to determine relevant biological pathways and processes and the Disease Ontology database²² (DOSE package in R) to examine the association of protein biomarkers of BP traits with other comorbidities.

We conducted 2-sample MR (TwoSampleMR package in R) to infer causal relations of plasma proteins to BP traits, using *cis*-pQTL variants as the instrumental variables. The *cis*-pQTL signals were generated from analysis of the UKB-Pharma Proteomics Project study.¹⁴ Instruments for each protein were selected by pruning the single nucleotide polymorphisms (SNPs) for linkage disequilibrium ($r^2 < 0.01$) using the European reference sample within the 1000 Genomes panel.²³ Postlinkage disequilibrium clumping, we explored causal relations of 2007 plasma proteins with SBP/DBP and 2018 proteins with hypertension. The results of GWAS of SBP and DBP¹³ and hypertension were used to derive the summary estimates for the outcomes. Among the 2538 Olink HT proteins measured exclusively in FHS, we performed MR for SBP/DBP (1235 proteins) and hypertension (1251 proteins), using the *cis*-pQTL signals from 5774 FHS participants as the instrumental variables. The inverse variance weighted method was used to perform multi-SNP MR, whereas single-SNP MR was performed using Wald ratio, with a significance threshold of FDR < 0.05 . Furthermore, we conducted sensitivity analysis for MR, where we evaluated the average pleiotropic effect of the SNPs by examining the MR-Egger intercept. A significance level of $P < 0.05$ indicated a nonzero MR-Egger intercept, signaling a violation of the no-pleiotropy assumption in MR. Instrument strength was evaluated using the F statistic, with all selected instrumental variables exceeding the conventional threshold of $F > 10$. Proteins significant in the MR were further tested for colocalization (coloc package in R) of their *cis*-pQTL signals with the GWAS signals of BP traits. Statistical significance for genetic colocalization was determined by the posterior probability of hypothesis 4 > 0.8 .

Hypothesis-4 assumes that both exposure (plasma proteins) and outcome (BP trait) share a common associated genetic signal. We evaluated colocalization of the genetic signals in 2 stages. First, we tested whether the plasma-protein pQTL and BP trait GWAS are consistent with a single shared causal variant assumption.²⁴ We then reanalyzed each locus with Sum of Single Effects fine-mapping,²⁵ which relaxes the single-variant assumption and accommodates multiple causal variants within the same region, recording the maximum posterior probability of hypothesis 4 per locus. Proteins whose loci achieved a posterior probability of hypothesis 4 > 0.80 in either the single- or multi-variant analysis were declared genetically colocalized with the corresponding BP trait.

A comprehensive description of the methods is available in the [Supplemental Material](#).

RESULTS

The UKB discovery sample consisted primarily (93%) of European ancestry participants. The remaining 7% of the sample was derived from several non-European ancestries (Table 1). Of the 45991 UKB participants included in the cross-sectional analysis, 54% were hypertensive at baseline. Individuals with hypertension were older, had a higher proportion of men, and exhibited reduced kidney function compared with the normotensive group. The proportion of former smokers was higher in the hypertensive group, whereas the normotensive group had a greater proportion of current smokers.

Association Between Plasma Proteins and BP Traits in UKB

In the UKB cross-sectional analysis, conventional linear and logistic regression yielded highly inflated *P* values, as indicated by the genomic inflation factor (λ_{GC} ; SBP=79, DBP=129, and hypertension=72). In contrast, λ_{GC} was 1.6 for the prospective analysis of new-onset hypertension (Figure S1A through S1D). These values closely matched those calculated (λ_{GC} ; SBP=78, DBP=130, and hypertension=42) from UKB association results published by Xu et al¹⁰ (Figure S1E through S1H). Conventional association analysis resulted in over 60% of the proteins analyzed being significant ($P < 0.000017$), with 1785 proteins associated with SBP, 1971 with DBP, and 1753 with hypertension at baseline (Tables S1 through S3). Applying a knockoff filter¹⁰ reduced the significant (FDR_{knockoff} < 0.1) set to 342, 276, and 200 proteins for SBP, DBP, and prevalent hypertension, respectively (Table 2 and Tables S4 through S6). In sensitivity analyses, 293 proteins associated with SBP remained significant after medication adjustment among treated participants (N=10 203), and 313 remained significant after additional adjustment for BMI (Figure S2 and Tables S7 and S8). For DBP, the corresponding numbers were 219 and 231, respectively (Figure S3; Tables S9 and S10). In sensitivity analysis of hypertension, 100

Table 1. Baseline Clinical Characteristics of 45 991 UK Biobank and 5760 Framingham Heart Study Participants

Variable	UK Biobank				Framingham Heart Study			
	All	Normotensive	Hypertensive	P value	All	Normotensive	Hypertensive	P value
N	45 991	21 239	24 752	...	5760	3583	2177	...
Age, y	56.8 (8.2)	53.9 (8.2)	59.3 (7.4)	<2×10 ⁻¹⁶	54.9 (13.5)	49.9 (11.9)	63.0 (11.8)	<2×10 ⁻¹⁶
Female sex	24 754 (53.8%)	13 070 (61.5%)	11 684 (47.2%)	<2×10 ⁻¹⁶	3135 (54.4%)	2078 (58.0%)	1057 (48.6%)	3.7×10 ⁻¹²
Ancestry								
European	42 792 (93.0%)	19 709 (92.8%)	23 083 (93.3%)	0.0551	5190 (90.1%)	3261 (91.0%)	1929 (88.6%)	0.0035
African	1090 (2.4%)	448 (2.1%)	642 (2.6%)	0.0007	145 (2.5%)	49 (1.4%)	96 (4.4%)	1.6×10 ⁻¹²
Asian	1015 (2.2)	495 (2.4%)	520 (2.1%)	0.1009	134 (2.3%)	84 (2.3%)	50 (2.3%)	0.9791
Hispanic	...	...	...	...	194 (3.4%)	111 (3.1%)	83 (3.8%)	0.1669
Mixed	316 (0.7%)	195 (0.9%)	121 (0.5%)	3.8×10 ⁻⁸	75 (1.3%)	62 (1.7%)	13 (0.6%)	0.0004
Others	778 (1.7%)	392 (1.8%)	386 (1.6%)	0.0195	22 (0.4%)	16 (0.4%)	6 (0.3%)	0.424
Smoking status								
Current	4829 (10.5%)	2540 (12.0%)	2289 (9.2%)	<2×10 ⁻¹⁶	587 (10.2%)	404 (11.3%)	183 (8.4%)	0.0006
Former	16 081 (35.0%)	6598 (31.1%)	9483 (38.3%)	<2×10 ⁻¹⁶	2127 (36.9%)	1132 (31.6%)	995 (45.7%)	<2×10 ⁻¹⁶
BMI, kg/m ²	27.4 (4.7)	26.1 (4.2)	28.5 (4.9)	<2×10 ⁻¹⁶	28.0 (5.6)	26.9 (5.0)	30.0 (6.0)	<2×10 ⁻¹⁶
Systolic BP, mmHg	137.6 (18.6)	124.0 (9.9)	150.0 (16.0)	<2×10 ⁻¹⁶	121.3 (16.6)	114.6 (11.6)	132.4 (17.6)	<2×10 ⁻¹⁶
Diastolic BP, mmHg	82.1 (10.2)	76.2 (7.1)	87.1 (9.7)	<2×10 ⁻¹⁶	73.9 (9.6)	72.6 (8.1)	76.1 (11.3)	<2×10 ⁻¹⁶
eGFR	94.1 (15.5)	97.8 (13.7)	90.9 (16.2)	<2×10 ⁻¹⁶	81.0 (16.8)	84.0 (15.0)	76.1 (18.4)	<2×10 ⁻¹⁶
Alcohol consumption, ounce/mo	14.1 (18.1)	12.4 (15.6)	15.5 (19.9)	<2×10 ⁻¹⁶	8.8 (14.0)	8.3 (1.24)	9.8 (16.3)	<2×10 ⁻¹⁶

Data are presented as No. (%) or mean (SD) unless otherwise indicated. The difference among normotensive and hypertensive groups were assessed using Student *t* test for the continuous variables and the χ^2 test for categorical variables. Hypertensive=defined as BP <140/90 or current use of antihypertensive medication. BMI indicates body mass index; BP, blood pressure; and eGFR, estimated glomerular filtration rate.

protein signals were confirmed when elevated BP was defined as SBP/DBP ≥130/80 mmHg, and 168 proteins remained associated after additionally adjusting for BMI (Figure S4; Tables S11 and S12). After removing participants who were receiving antihypertensive treatment, 104 associations with hypertension persisted, with the direction of effect for 11 proteins—most notably renin—reversed (Table S13). Prospective analysis identified 60 proteins associated with new-onset hypertension; after adjustment for BMI, only renin, leptin, and oligodendrocyte-myelin glycoprotein (OMG) remained significant (Tables S14 and S15).

Validation of Protein Biomarkers of BP Traits in FHS

A total of 329 (out of 342 that were significant in the UKB discovery), 262 (of 276), and 193 (of 200) proteins—in the discovery analysis of SBP, DBP, and prevalent hypertension were available for validation in FHS. Of these, 115 proteins for SBP, 109 for DBP, and 102 for hypertension were validated using a Bonferroni-corrected significance threshold (Figure S5; Tables S4 through S6), while a total of 184 proteins for SBP, 166 for DBP, and 142 for hypertension reached nominal significance (*P*<0.05). Of the 56 proteins significant in UKB and available for validation in FHS for new-onset hypertension, 17 proteins were

validated and overall, 34 proteins reached nominal significance (Table S14).

Underlying Biological Mechanisms of Protein Biomarkers Associated With BP Traits

Enrichment analyses used 276, 342, and 200 proteins significantly associated with SBP, DBP, and hypertension, respectively, in discovery analyses as the target, and all the 2922 proteins from UKB as the background. These analyses revealed that many of the proteins associated with hypertension and BP are involved in blood vessel and ventricular cardiac muscle tissue morphogenesis, vasoconstriction, and both proliferation and apoptosis of epithelial cells (Figure 2A through 2C and Table S16). These proteins also participate in positive regulation of the MAPK (mitogen-activated protein kinase) pathway and immune responses such as inflammation. Furthermore, these proteins are more likely to be associated with cardiovascular, renal, and hepatobiliary disorders, as well as rheumatic diseases (Figure 3; Figure S6; Table S17).

Causal Inference Analysis

Among the proteins that were significant in the cross-sectional analysis in the UKB, MR identified 37 proteins with putative causal associations with SBP, 25 with DBP,

Table 2. Putatively Causal Biomarkers of SBP, DBP, and Hypertension Identified by MR, Association Analysis, and Genetic Colocalization

Uniprot ID	Protein	Gene	Phenotype	Direction of effect			Colocalized signal
				MR	Knockoff/ elastic-net	Linear regression	
P12821	Angiotensin-converting enzyme	ACE	Hypertension	Positive	Positive	Positive	Yes
P02647	Apolipoprotein A-I	APOA1	DBP	Positive	Positive	Positive	Yes
P02649	Apolipoprotein E	APOE	Hypertension	Negative	Negative	NS	Yes
Q9HD42	Charged multivesicular body protein 1a	CHMP1A	SBP	Positive	Negative	Positive	Yes
P12034	Fibroblast growth factor 5	FGF5	SBP	Positive	Positive	NS	Yes
P12034	Fibroblast growth factor 5	FGF5	DBP	Positive	Positive	Negative	Yes
P12034	Fibroblast growth factor 5	FGF5	Hypertension	Positive	Positive	NS	Yes
P02751	Fibronectin	FN1	SBP	Negative	Negative	Positive	Yes
P09958	Furin	FURIN	SBP	Positive	Negative	Positive	Yes
P17936	Insulin-like growth factor-binding protein 3	IGFBP3	DBP	Positive	Positive	Positive	Yes
Q96JA1	Leucine-rich repeats and immunoglobulin-like domains protein 1	LRIG1	SBP	Positive	Positive	Positive	Yes
Q14767	Latent-transforming growth factor beta-binding protein 2	LTBP2	SBP	Negative	Positive	NS	Yes
O95544	NAD kinase	NADK	SBP	Negative	Positive	Positive	Yes
P04054	Phospholipase A2	PLA2G1B	DBP	Negative	Negative	Negative	Yes
Q15599	Na(+)/h(+) exchange regulatory cofactor NHE-RF2	SLC9A3R2	SBP	Positive	Positive	Positive	Yes
Q15599	Na(+)/h(+) exchange regulatory cofactor NHE-RF2	SLC9A3R2	Hypertension	Positive	Positive	Positive	Yes
P35590	Tyrosine-protein kinase receptor Tie-1	TIE1	SBP	Negative	Negative	NS	Yes
P23381	Tryptophan-tRNA ligase, cytoplasmic	WARS1	SBP	Positive	Positive	Positive	Yes

DBP indicates diastolic blood pressure; MR, Mendelian randomization; and SBP, systolic blood pressure.

and 7 with hypertension (Figure 4). Of these, 10 proteins shared genetic signals with SBP, 4 with DBP, and 4 with hypertension (Table 2). In addition, 169, 185, and 75 proteins were significant in the MR but lacked significance in the UKB analysis of protein associations with SBP, DBP, and hypertension, respectively (Tables S18 through S20). Three proteins— FGF5 (fibroblast growth factor 5), ACE (angiotensin-converting enzyme), and NHERF2 (Na⁺/H⁺ exchange regulatory cofactor 2), encoded by the gene *SLC9A3R2*, were associated with SBP, DBP, and prevalent hypertension in both association and MR analyses. FGF5 exhibited shared genetic signals with all the 3 BP phenotypes.

Secondary Analyses of the Additional 2500 Proteins in FHS

This analysis identified 89 proteins associated ($P<9.4\times10^{-6}$) with SBP, 67 with DBP, and 162 with prevalent hypertension (Tables S21 through S23). Of these proteins, MR analyses provided evidence of putative causality for 6 proteins (29 total MR associations; FDR <0.05) with SBP, 4 with DBP (30 total), and 1 (12 total)—EDH4 (EH domain-containing protein 4)—with prevalent hypertension (Table S24). Three proteins—NECAB2 (N-terminal EF-hand calcium-binding protein 2), CTH (cystathionine gamma-lyase), and

HPD (4-hydroxyphenylpyruvate dioxygenase)—were prospectively associated with new-onset hypertension (Table S25).

DISCUSSION

In a proteome-wide screen of 2922 circulating proteins, we uncovered and independently validated a broad panel of protein biomarkers associated with SBP, DBP, and hypertension. Enrichment analysis highlighted the MAPK cascade as a pivotal driver of hypertension,²⁶ consistent with its roles in promoting inflammation,²⁷ endothelial dysfunction,²⁸ vascular remodeling,²⁹ and cardiac hypertrophy.³⁰ The same protein set was also overrepresented in cardiovascular, renal,³¹ hepatobiliary,³² and rheumatic disease³³ ontologies, suggesting common inflammatory and autoimmune axes linking hypertension to multi-organ comorbidity.³³

MR of 2007 proteins, including 1198 proteins not previously explored,¹⁰ uncovered novel putatively causal associations for 185 proteins with SBP and 194 with DBP. Integrating evidence from MR, genetic colocalization, and protein-trait association analyses, 15 plasma proteins were identified with strong support for a causal role in BP regulation and hypertension. Of these, 6 proteins—FN1 (fibronectin), LTBP2 (latent-transforming growth factor β -binding protein 2), NADK (NAD kinase),

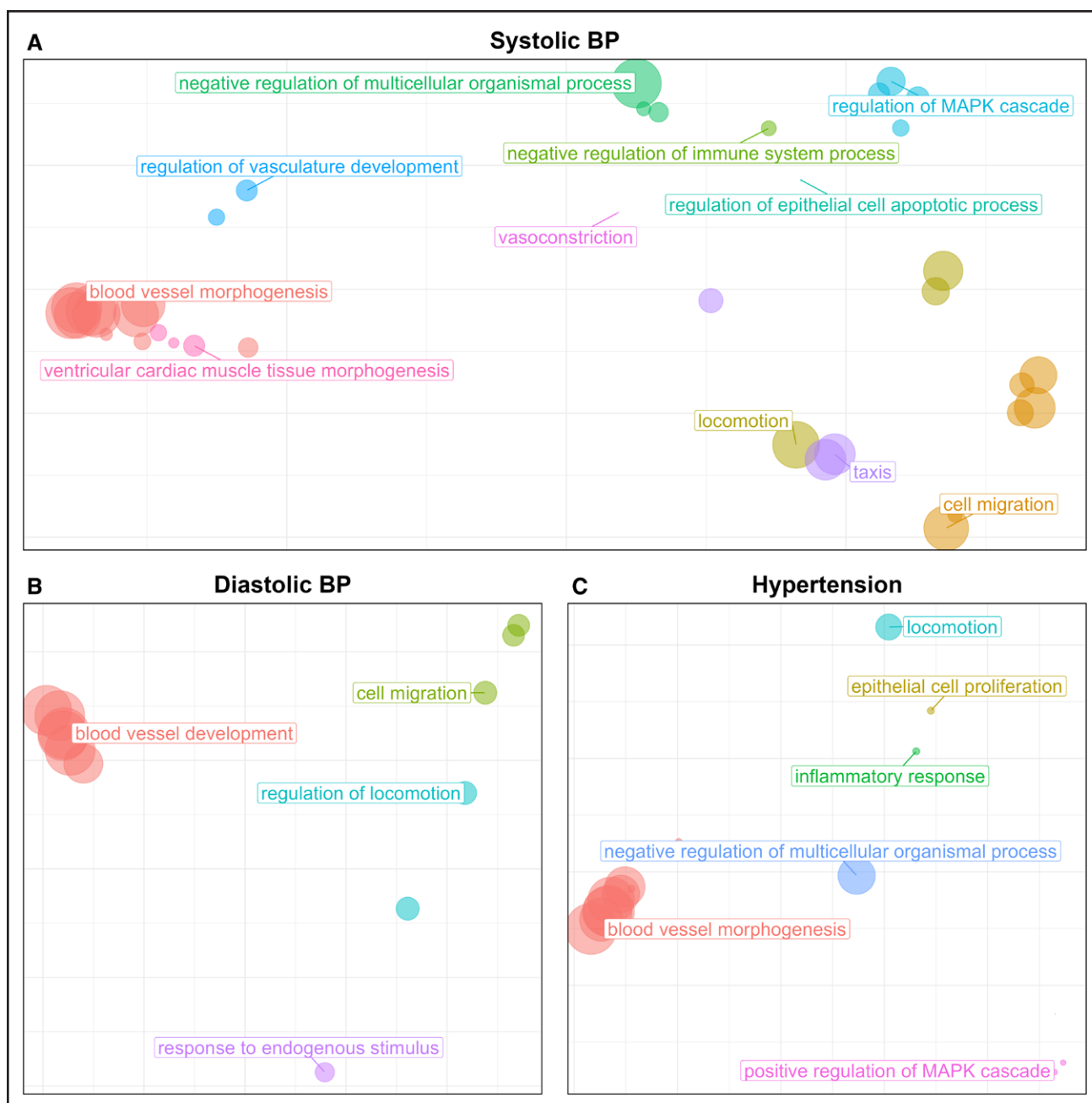


Figure 2. Enrichment of Gene Ontology (GO) biological process terms among proteins associated with systolic and diastolic blood pressure (BP), and hypertension.

This figure illustrates the most significantly enriched GO biological process terms identified from proteins associated with (A) systolic BP, (B) diastolic BP, and (C) hypertension. The figure highlights key biological processes implicated in BP regulation and hypertension pathogenesis. Each node represents a significant GO term (Bonferroni-adjusted $P < 0.05$). Only the parent term is labeled for all significant nodes. Shorter distances between nodes indicate greater similarity between GO terms, and larger node sizes correspond to higher statistical significance. MAPK indicates mitogen-activated protein kinase.

TIE1 (tyrosine-protein kinase receptor Tie-1), PLA2G1B (phospholipase A2), and APOE (apolipoprotein E)—were inversely associated with BP traits in MR, suggesting protective roles. LTBP2, TIE1, and PLA2G1B emerged as novel biomarkers associated with BP traits. MR associations of NADK with BP traits were also observed by Xu et al.¹⁰ TIE1 is an endothelial protein involved in angiogenesis, and its circulating levels have been

associated with decreased hypertension risk.³⁴ LTBP2 is linked to mitochondrial apoptosis, and its knockdown facilitates vascular remodeling,³⁵ suggesting a protective role in vascular disease. PLA2G1B is primarily known for its role in lipid metabolism and inflammation.³⁶ MR suggested an inverse association of NADK with SBP. In contrast, protein-trait association revealed a positive association of NADK with SBP, consistent with

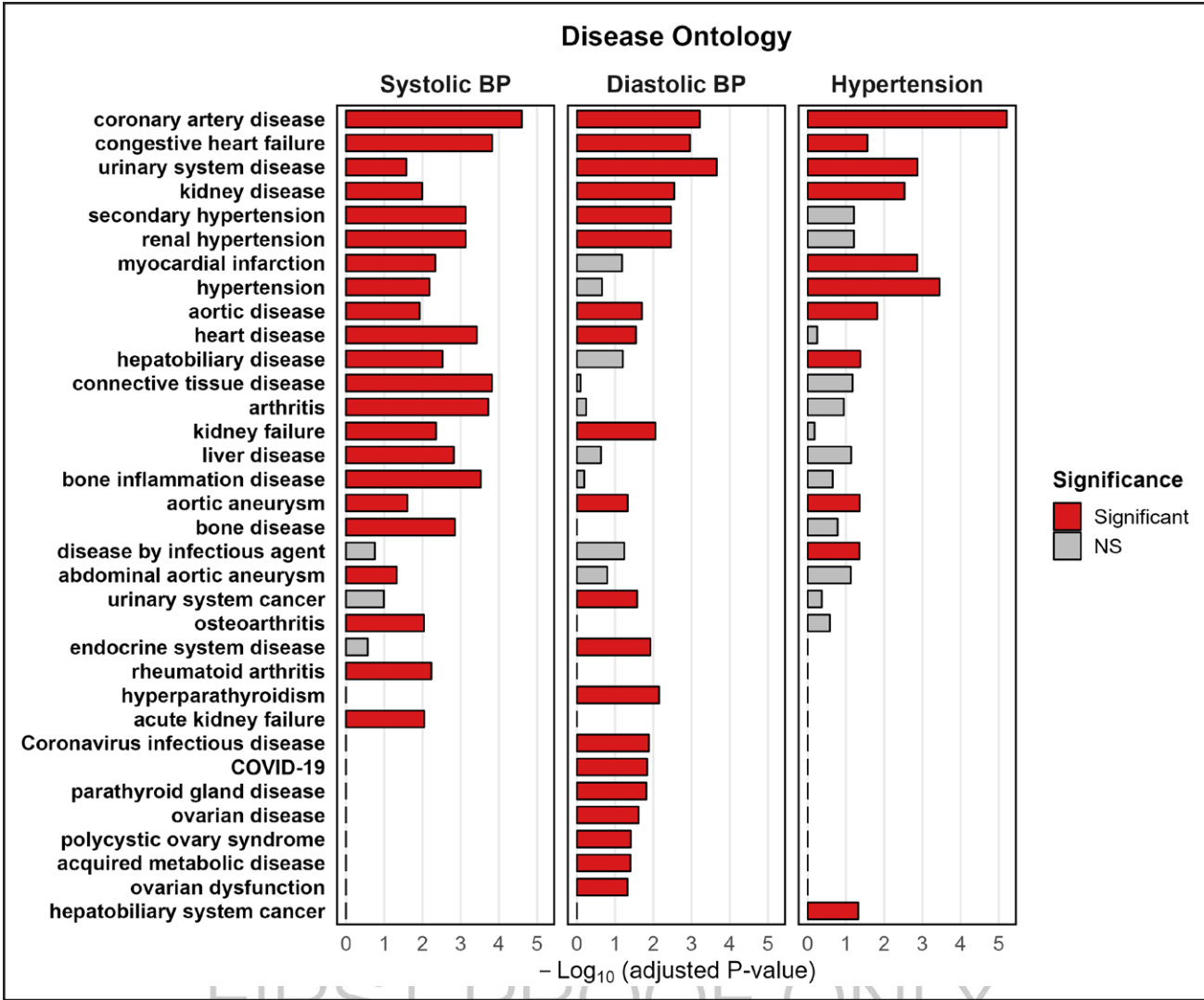


Figure 3. Disease Ontology (DO) profiles enriched among proteins associated with systolic and diastolic blood pressure (BP), and hypertension. The figure shows all DO terms (y axis) that are enriched among protein expression profiles associated with systolic BP, diastolic BP, and hypertension. The x axis shows the negative logarithm (base 10) of the adjusted P value (Bonferroni correction). DO terms with significant enrichment (Bonferroni-adjusted $P < 0.05$) are highlighted in red.

NADK-driven counterregulation in the setting of hypertension, which corresponds with previously observed declines in NAD⁺ levels, its substrate, in hypertension.³⁷ Association of APOE with BP may depend on the specific genotype: the APOE ε4 allele has been associated with higher BP;³⁸ while the ε2/ε3 genotype is associated with reduced hypertension risk.³⁹ Although FN1 showed an inverse association with BP in MR analysis, its plasma levels were positively associated with BP, consistent with previous observations.^{40–42} FN1 plays a key role in wound healing, and its prolonged activity in chronic conditions like hypertension may contribute to vascular⁴³ and cardiac remodeling.⁴⁴

Nine proteins—FGF5, NHERF2, ACE, furin, CHMP1A (charged multivesicular body protein 1A), LRIG1 (leucine-rich repeats and immunoglobulin-like domains protein 1),

WARS1 (tryptophan-tRNA ligase, cytoplasmic), APOA1 (apolipoprotein A-I), and IGFBP3 (insulin-like growth factor-binding protein 3)—were positively associated with BP traits in MR. ACE has a well-established role in elevating BP through the renin-angiotensin-aldosterone system. Furin can elevate BP by influencing the renin-angiotensin-aldosterone system⁴⁵ in addition to activating sodium channels.⁴⁶ The positive association between furin and BP in both cross-sectional and longitudinal analyses aligns with its biology and its reported association with cardiovascular disease risk,⁴⁷ despite prior reports of negative associations with BP.⁴⁸ FGF5 has been implicated in angiogenesis,^{49,50} and its locus has been genetically linked with BP in multiple GWAS.^{51,52} FGF5 demonstrated similar positive associations with BP traits in a 2018 study⁵³ and has more recently been

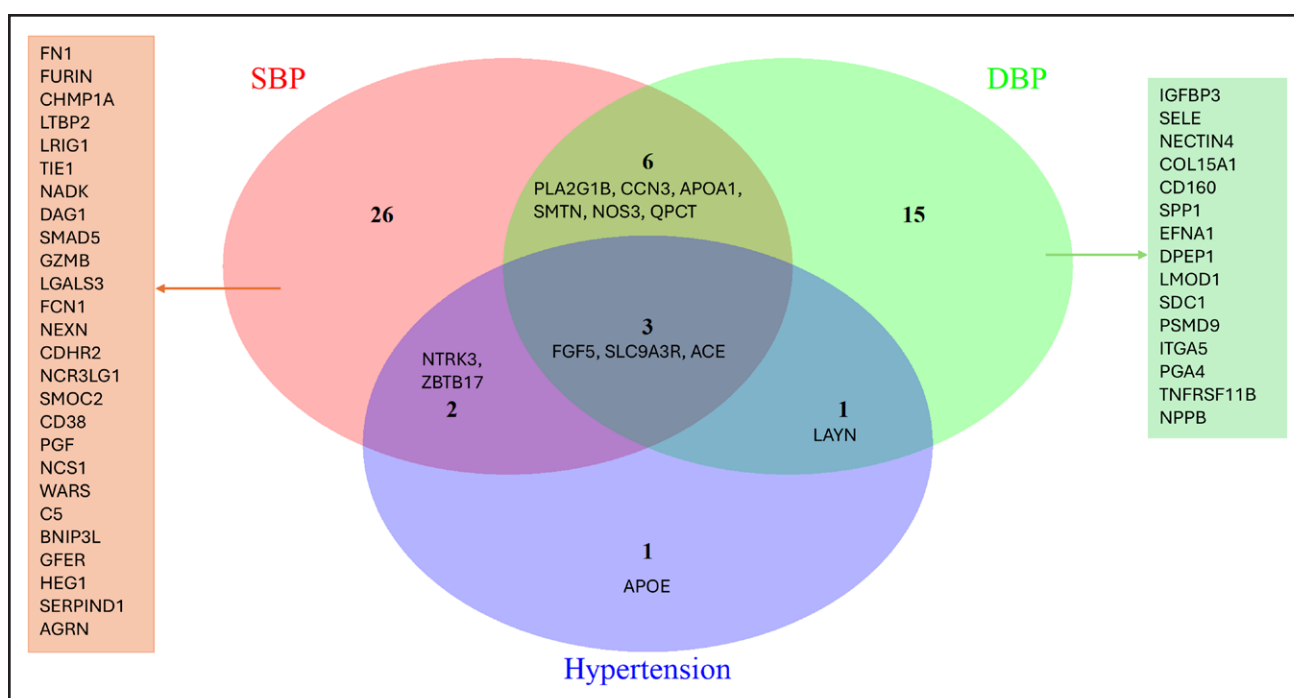


Figure 4. Putatively causal proteins identified by Mendelian randomization (MR) and association analyses.



proposed as a mediator of BP-driven cardiac remodeling.⁵⁴ APOA1, the principal component of high-density lipoprotein cholesterol, is generally considered cardioprotective and has been reported to be inversely associated with BP traits.⁵⁵ In contrast, we observed a positive association of APOA1 with DBP in both MR and protein-trait association analyses. This finding is consistent with emerging evidence that very high APOA1 levels may reflect dysfunctional high-density lipoprotein biology, with inflammation- and oxidation-related modifications that may impair vascular function and potentially contribute to increased vascular resistance.⁵⁶ The positive association of IGFBP3 with BP in MR analysis is consistent with an earlier observation.⁵⁷ NHERF2, WARS1, LRIG1, and CHMP1A were among the other novel biomarkers of BP traits that we identified in this study.

NHERF2, encoded by *SLC9A3R2*, emerged as a high-confidence novel target with MR suggesting a causal link with all BP phenotypes and with new-onset hypertension (Table S14). Large-scale genetic studies have identified rare loss-of-function and deleterious missense variants in *SLC9A3R2* that confer protection against hypertension.^{58,59} NHERF2 interacts with Na⁺/H⁺ exchanger 3, a primary sodium transporter in the kidney and intestine, thereby modulating sodium absorption and influencing BP regulation.^{60,61} NHERF2 could represent a promising therapeutic target for lowering BP. WARS1 triggers inflammation,⁶² and a recently identified variant associated with splice variation of *WARS1* has been associated with elevated plasma WARS1 levels and increased hypertension risk.⁶³ LRIG1 regulates lipid metabolism⁶⁴ and may also play a role in suppressing inflammation.⁶⁵

The only link of CHMP1A to hypertension comes from a proteomics study of kidney tissue, where CHMP1A was linked with chronic kidney disease and hypertension,⁶⁶ suggesting a shared mechanism.

Leptin was a significant predictor of new-onset hypertension, independent of BMI, suggesting a pressor effect beyond its association with adiposity. Experimental blocking of leptin signaling has been shown to lower BP, and individuals with leptin signaling defects are protected from obesity-related hypertension.⁶⁷ Conversely, lower baseline renin—paralleling its inverse cross-sectional association with SBP and DBP—predicted new-onset hypertension. Recent prospective studies likewise show that low renin is associated with risk of hypertension, particularly in salt-sensitive hypertension or when the aldosterone-to-renin ratio is elevated.^{68,69} When individuals on antihypertensive therapy were excluded, renin levels were also lower in individuals with preexisting hypertension, indicating that the opposite direction of effect in the full analysis is likely driven by medication effects on renin levels (Table S13). In secondary analysis of 2538 proteins that were available only in FHS, VTN (vitronectin) and EHD4, both proteins known to facilitate angiogenesis and vascular remodeling,^{70,71} were identified as promising biomarkers.

Our study has several major strengths. To our knowledge, it represents the largest proteomic investigation with an integrative design focused on biomarkers of BP and hypertension, complemented by independent external validation, reflecting the scientific rigor of our findings. In addition, we implemented a rigorous statistical method to control for inflation while analyzing thousands

of highly intercorrelated plasma proteins, yielding more robust and credible associations. Furthermore, we leveraged the most recent and expansive GWAS data sets for both exposure and outcome, providing more reliable genetic instruments and enhancing the credibility of the results of MR and colocalization analyses. We also conducted sensitivity analyses to rule out potential horizontal pleiotropy in our MR results, further strengthening our conclusions.

Our study also has several limitations. As UKB is a volunteer-based cohort, nonrandom participation may limit generalizability and could influence observed protein-trait associations.⁷² In addition, the study sample consisted predominantly of participants of European ancestry, limiting the generalizability of our findings to other racial or ethnic groups, although 7% of participants in the discovery and $\approx 10\%$ of those in the validation cohort were of non-European ancestries. Another limitation is the overlap of participants in the sample used to derive the GWAS summary estimates for both exposure and outcome in our 2-sample MR, which could have introduced bias in our MR results. Nonetheless, by utilizing the largest available GWAS resource, we ensured the robustness of our genetic instruments—an essential requirement for 2-sample MR. Furthermore, because our proteomic data lack isoform-level resolution, our results may obscure isoform-specific effects on BP regulation and hypertension. Lastly, our prospective analysis suffered from reduced power due to a smaller sample size of participants with follow-up BP data, which limited our ability to discover proteins associated with new-onset hypertension.

In summary, our findings shed new light on the biological underpinnings of hypertension and point to concrete opportunities for precision diagnostics and therapeutics. We identified several novel, putatively causal protein biomarkers that might influence BP regulation and the development of hypertension, each representing a potential therapeutic target. However, because these conclusions are derived from integrative genomic approaches such as MR and genetic colocalization, the inferred causal associations remain speculative. Clinical validation of these biomarkers—especially in racially and ethnically diverse populations—is essential to confirm their relevance and ensure their broad applicability. Additional studies are warranted to elucidate their mechanisms of action and evaluate their clinical utility.

PERSPECTIVES

This study marks a significant advance in disease biomarker research, providing novel insights into the proteomic architecture of BP regulation. We applied the innovative reflection knockoff method to rigorously control the FDR to identify plasma-protein biomarkers of BP traits using a high-dimensional proteomic data set. By

leveraging a comprehensive proteomic panel and the knockoff strategy in a large UKB sample, our analysis achieved a greater statistical power and more precise FDR control compared with previous efforts. This enabled us to discover a replicable set of protein biomarkers associated with BP and hypertension, which was further supported by downstream enrichment analysis showing significant overlap with established hypertension-related biological processes and comorbidity disease terms. In addition, we utilized the most up-to-date and largest GWAS summary data sets for the MR and genetic colocalization analyses, expanding the list of promising candidate protein targets for therapeutic intervention. These findings lay the groundwork for future clinical studies where these biomarkers could be validated to develop improved therapeutics to manage hypertension.

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Disclosures

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REFERENCES

1. NCD Risk Factor Collaboration (NCD-RisC). Worldwide trends in blood pressure from 1975 to 2015: a pooled analysis of 1479 population-based measurement studies with 19.1 million participants. *Lancet*. 2017;389:37–55. doi: 10.1016/S0140-6736(16)31919-5
2. *Hypertension*. <https://www.who.int/news-room/fact-sheets/detail/hypertension>
3. Davis BR, Hennekens CH, Greenwald J, Dunn J, Matarazzo A, Benson K, Kitasantas P, Rubin S. New clinical challenges in hypertension management: the “old silent killer” is alive and well. *Am J Med*. 2024;137:1154–1156. doi: 10.1016/j.amjmed.2024.07.005

4. Abdalla M, Bolen SD, Brettler J, Egan BM, Ferdinand KC, Ford CD, Lackland DT, Wall HK, Shimbo D; American Heart Association and American Medical Association. Implementation strategies to improve blood pressure control in the United States: a scientific statement from the American Heart Association and American Medical Association. *Hypertension*. 2023;80:e143–e157. doi: 10.1161/HYP.0000000000000232
5. Flack JM. Better implementation of what is known will reverse falling hypertension control rates: we know how to reverse falling hypertension rates. *Hypertension*. 2021;78:588–590. doi: 10.1161/HYPERTENSIONAHA.121.17512
6. Wang TJ, Gona P, Larson MG, Levy D, Benjamin EJ, Tofler GH, Jacques PF, Meigs JB, Rifai N, Selhub J, et al. Multiple biomarkers and the risk of incident hypertension. *Hypertension*. 2007;49:432–438. doi: 10.1161/01.HYP.0000256956.61872.a
7. Charkiewicz AE, Garley M, Ratajczak-Wrona W, Jabłońska E, Mityk W, Motyka J, Omeljanuk WJ. The diagnostic potential of novel biomarkers of hypertension in men. *Arch Med Sci*. 2022;18:1666–1671. doi: 10.5114/aoms/153500
8. Hidru TH, Yang X, Xia Y, Ma L, Li HH. The relationship between plasma markers and essential hypertension in middle-aged and elderly chinese population: a community based cross-sectional study. *Sci Rep*. 2019;9:6813. doi: 10.1038/s41598-019-43278-4
9. Harrison DG, Coffman TM, Wilcox CS. Pathophysiology of hypertension. *Circ Res*. 2021;128:847–863. doi: 10.1161/CIRCRESAHA.121.318082
10. Xu S, Wen S, Zong X, Wen S, Zhu J, Zheng W, Wang Z, Cao P, Liang Z, Ding C, et al. Identification of circulating proteins associated with blood pressure. *Hypertension*. 2025;82:333–346. doi: 10.1161/HYPERTENSIONAHA.124.24151
11. Micheel CM, Nass SJ, Omenn GS, editors. *Evolution of Translational Omics: Lessons Learned and the Path Forward*. National Academies Press; 2012. <http://www.nap.edu/catalog/13297>
12. Guan Y, Levy D. Reflection knockoffs via householder reflection: applications in proteomics and genetic fine mapping. *Genetics*. 2025;231:iyaf178. doi: 10.1093/genetics/iyaf178
13. Keaton JM, Kamali Z, Xie T, Vaez A, Williams A, Goleva SB, Ani A, Evangelou E, Hellwege JN, Yengo L, et al; Million Veteran Program. Genome-wide analysis in over 1 million individuals of European ancestry yields improved polygenic risk scores for blood pressure traits. *Nat Genet*. 2024;56:778–791. doi: 10.1038/s41588-024-01714-w
14. Sun BB, Chiou J, Traylor M, Benner C, Hsu Y-H, Richardson TG, Surendran P, Mahajan A, Robins C, Vasquez-Grinnell SG, et al; Alnylam Human Genetics. Plasma proteomic associations with genetics and health in the UK Biobank. *Nature*. 2023;622:329–338. doi: 10.1038/s41586-023-06592-6
15. *Olink Explore HT*. *Olink*®. <https://olink.com/products/olink-explore-hi>
16. *Olink Explore 3072/384*. *Olink*®. <https://olink.com/products/olink-explore-3072-384>
17. Inker LA, Eneanya ND, Coresh J, Tighiouart H, Wang D, Sang Y, Crews DC, Doria A, Estrella MM, Froissart M, et al; Chronic Kidney Disease Epidemiology Collaboration. New creatinine- and cystatin C-based equations to estimate GFR without race. *N Engl J Med*. 2021;385:1737–1749. doi: 10.1056/NEJMoa2102953
18. Levey AS, Inker LA, Coresh J. GFR estimation: from physiology to public health. *Am J Kidney Dis*. 2014;63:820–834. doi: 10.1053/j.ajkd.2013.12.006
19. Tobin MD, Sheehan NA, Scurrah KJ, Burton PR. Adjusting for treatment effects in studies of quantitative traits: antihypertensive therapy and systolic blood pressure. *Stat Med*. 2005;24:2911–2935. doi: 10.1002/sim.2165
20. Shihab HM, Meoni LA, Chu AY, Wang NY, Ford DE, Liang K-Y, Gallo JJ, Klag MJ. Body mass index and risk of incident hypertension over the life course. *Circulation*. 2012;126:2983–2989. doi: 10.1161/CIRCULATIONAHA.112.117333
21. Aleksander SA, Balhoff J, Carbone S, Cherry JM, Drabkin HJ, Ebert D, Feuermann M, Gaudet P, Harris NL, Hill DP, et al; Gene Ontology Consortium. The Gene Ontology knowledgebase in 2023. *Genetics*. 2023;224:iyad031. doi: 10.1093/genetics/iyad031
22. Schriml LM, Arze C, Nadendla S, Chang YWW, Mazaitis M, Felix V, Feng G, Kibbe WA. Disease ontology: a backbone for disease semantic integration. *Nucleic Acids Res*. 2012;40:D940–D946. doi: 10.1093/nar/gkr972
23. McVean GA, Altshuler DM, Durbin RM, Abecasis GR, Bentley DR, Chakravarti A, Clark AG, Donnelly P, Eichler EE, Flicke P, et al. An integrated map of genetic variation from 1,092 human genomes. *Nature*. 2012;491:56–65. doi: 10.1038/nature11632
24. Giambartolomei C, Vukcevic D, Schadt EE, Franke L, Hingorani AD, Wallace C, Plagnol V. Bayesian test for colocalisation between pairs of genetic association studies using summary statistics. *PLoS Genet*. 2014;10:e1004383. doi: 10.1371/journal.pgen.1004383
25. Wallace C. A more accurate method for colocalisation analysis allowing for multiple causal variants. *PLoS Genet*. 2021;17:e1009440. doi: 10.1371/journal.pgen.1009440
26. Ng GYQ, Loh ZW-L, Fann DY, Mallilankaraman K, Arumugam TV, Hande MP. Role of Mitogen-Activated Protein (MAP) kinase pathways in metabolic diseases. *Genome Integr*. 2024;15:e20230003. doi: 10.14293/genint.14.1.004
27. Moens U, Kostenko S, Sveinbjörnsson B. The Role of Mitogen-Activated Protein Kinase-Activated Protein Kinases (MAPKAPKs) in inflammation. *Genes (Basel)*. 2013;4:101–133. doi: 10.3390/genes4020101
28. Ramalingam P, Poulos MG, Lazzari E, Gutkin MC, Lopez D, Kloss CC, Crowley MJ, Katsnelson L, Freire AG, Greenblatt MB, et al. Chronic activation of endothelial MAPK disrupts hematopoiesis via NFkB dependent inflammatory stress reversible by SCGF. *Nat Commun*. 2020;11:666. doi: 10.1038/s41467-020-14478-8
29. Wang X, Liu R, Liu D. The role of the MAPK signaling pathway in cardiovascular disease: pathophysiological mechanisms and clinical therapy. *Int J Mol Sci*. 2025;26:2667. doi: 10.3390/ijms26062667
30. Streicher JM, Ren S, Herschman H, Wang Y. MAPK-activated protein kinase-2 in cardiac hypertrophy and cyclooxygenase-2 regulation in heart. *Circ Res*. 2010;106:1434–1443. doi: 10.1161/CIRCRESAHA.109.213199
31. Burnier M, Damianaki A. Hypertension as cardiovascular risk factor in chronic kidney disease. *Circ Res*. 2023;132:1050–1063. doi: 10.1161/CIRCRESAHA.122.321762
32. Viktorovna SE, Alekseevich NY, Yakovlevich PV, Michailovich MI. Association of arterial hypertension with hepatobiliary pathology: the occurrence of comorbidity and features of metabolic processes. *Curr Hypertens Rev*. 2020;16:138–147. doi: 10.2174/1573402115666190801104227
33. Taylor EB, Wolf VL, Dent E, Ryan MJ. Mechanisms of hypertension in autoimmune rheumatic diseases. *Br J Pharmacol*. 2019;176:1897–1913. doi: 10.1111/bph.14604
34. Li J, Yao Y-X, Yao P-S. Circulating biomarkers and risk of hypertension: a two-sample Mendelian randomisation study. *Heart Lung Circ*. 2023;32:1434–1442. doi: 10.1016/j.hlc.2023.09.003
35. Zhou C, Sun J, Wu L, Liu C, Cheng Q, Xie S, Zhang J. LTBP2 down-regulated FGF2 to repress vascular smooth muscle cell proliferation and vascular remodeling in a rat model of intracranial aneurysm. *Neurosci Lett*. 2024;842:137988. doi: 10.1016/j.neulet.2024.137988
36. Guan S, Xu Z, Yang T, Zhang Y, Zheng Y, Chen T, Liu H, Zhou J. Identifying potential targets for preventing cancer progression through the PLA2G1B recombinant protein using bioinformatics and machine learning methods. *Int J Biol Macromol*. 2024;276:133918. doi: 10.1016/j.ijbiomac.2024.133918
37. Qiu Y, Xu S, Chen X, Wu X, Zhou Z, Zhang J, Tu Q, Dong B, Liu Z, He J, et al. NAD⁺ exhaustion by CD38 upregulation contributes to blood pressure elevation and vascular damage in hypertension. *Signal Transduct Target Ther*. 2023;8:353. doi: 10.1038/s41392-023-01577-3
38. Leã Correa Leite M, Fernanda Lima-Costa M, Moriguchi EH. Age-related trends of blood pressure levels by apolipoprotein E genotype: the Bambuí cohort study of ageing (1997–2008). *Hypertens Res*. 2013;36:270–276. doi: 10.1038/hr.2012.175
39. Kuo CL, Pilling LC, Atkins JL, Kuchel GA, Melzer D. ApoE e2 and aging-related outcomes in 379,000 UK Biobank participants. *Aging (Albany NY)*. 2020;12:12222–12233. doi: 10.18632/aging.103405
40. Bizbiz L, Labat-Robert J, Alperovitch A, Robert L; EVA-group. Cardiovascular determinants of plasma fibronectin in an elderly population: the EVA study. *Arch Gerontol Geriatr*. 1997;25:201–209. doi: 10.1016/s0167-4943(97)00011-3
41. Han MS, Lee KW, Hwang SG, Park JD, Park SW. A study of the relationship of preeclampsia with total plasma fibronectin. *Korean J Obstet Gynecol*. 2001;44:1024–1029.
42. Song KS, Kim HK, Shim W, Jee SH. Plasma fibronectin levels in ischemic heart disease. *Atherosclerosis*. 2001;154:449–453. doi: 10.1016/s0021-9150(00)00490-1
43. Shi X, Zhang S, Li J, Ke Y, Bai Y. Fibronectin/ α 5 integrin contribute to hypertension-associated arterial ageing and calcification through affecting BMP2/MGP imbalance and enhancing vascular smooth muscle cell phenotypic transformation. *Gerontology*. 2024;70:858–875. doi: 10.1159/000539399
44. Valiente-Alandi I, Potter SJ, Salvador AM, Schafer AE, Schips T, Carrillo-Salinas F, Gibson AM, Nieman ML, Perkins C, Sargent MA, et al. Inhibiting fibronectin attenuates fibrosis and improves cardiac function in a model of heart failure. *Circulation*. 2018;138:1236–1252. doi: 10.1161/CIRCULATIONAHA.118.034609
45. Gonzalez AA, Lara LS, Luffman C, Seth DM, Prieto MC. The soluble form of the (pro)renin receptor [s(P)RR] is augmented in the collecting duct and

- urine of chronic angiotensin II-dependent hypertensive rats. *Hypertension*. 2011;57:859–864. doi: 10.1161/HYPERTENSIONAHA.110.167957
46. Zheng H, Helms MN, Zou C, Zimmerman E, Feng Y, Yang T. Furin-mediated modification is required for epithelial sodium channel-activating activity of soluble (pro)renin receptor in cultured collecting duct cells. *Am J Physiol Renal Physiol*. 2025;328:F766–F774. doi: 10.1152/ajprenal.00087.2024
 47. Fry H, Mazidi M, Kartsonaki C, Clarke R, Walters RG, Chen Z, Millwood IY. The role of furin and its therapeutic potential in cardiovascular disease risk. *Int J Mol Sci*. 2024;25:9237. doi: 10.3390/ijms25179237
 48. He Y, Ren L, Zhang Q, Zhang M, Shi J, Hu W, Peng H, Zhang Y. Serum furin as a biomarker of high blood pressure: findings from a longitudinal study in Chinese adults. *Hypertens Res*. 2019;42:1808–1815. doi: 10.1038/s41440-019-0295-6
 49. Seo HR, Jeong HE, Joo HJ, Choi SC, Park CY, Kim JH, Choi JH, Cui LH, Hong SJ, Chung S, et al. Intrinsic FGF2 and FGF5 promotes angiogenesis of human aortic endothelial cells in 3D microfluidic angiogenesis system. *Sci Rep*. 2016;6:28832. doi: 10.1038/srep28832
 50. Giordano FJ, Ping P, McKinnan MD, Nozaki S, DeMaria AN, Dillmann WH, Mathieu-Costello O, Hammond HK. Intracoronary gene transfer of fibroblast growth factor-5 increases blood flow and contractile function in an ischemic region of the heart. *Nat Med*. 1996;2:534–539. doi: 10.1038/nm0596-534
 51. Newton-Cheh C, Johnson T, Gateva V, Tobin MD, Bochud M, Coin L, Najjar SS, Zhao JH, Heath SC, Eyheramendy S, et al; Wellcome Trust Case Control Consortium. Eight blood pressure loci identified by genome-wide association study of 34,433 people of European ancestry. *Nat Genet*. 2009;41:666–676. doi: 10.1038/ng.361
 52. Liu C, Li H, Qi Q, Lu L, Gan W, Loos RJ, Lin X. Common variants in or near FGF5, CYP17A1 and MTHFR genes are associated with blood pressure and hypertension in Chinese Hans. *J Hypertens*. 2011;29:70–75. doi: 10.1097/HJH.0b013e32833f60ab
 53. Ren Y, Jiao X, Zhang L. Expression level of Fibroblast Growth Factor 5 (FGF5) in the peripheral blood of primary hypertension and its clinical significance. *Saudi J Biol Sci*. 2018;25:469–473. doi: 10.1016/j.sjbs.2017.11.043
 54. Hu Y, Lin L, Zhang L, Li Y, Cui X, Lu M, Zhang Z, Guan X, Zhang M, Hao J, et al. Identification of circulating plasma proteins as a mediator of hypertension-driven cardiac remodeling: a Mediation mendelian randomization study. *Hypertension*. 2024;81:1132–1144. doi: 10.1161/HYPERTENSIONAHA.123.22504
 55. Dong H, Yang X, Zhang Y, Hu P, Liu Y, Liang S. Associations of serum apolipoprotein A1, B levels and their ratio with blood pressure in Chinese adults with coronary artery disease. *Blood Press Monit*. 2021;26:401–406. doi: 10.1097/MBP.0000000000000549
 56. Faaborg-Andersen CC, Liu C, Subramaniam V, Desai SR, Sun YV, Wilson PWF, Sperling LS, Quyyumi AA. U-shaped relationship between apolipoprotein A1 levels and mortality risk in men and women. *Eur J Prev Cardiol*. 2023;30:293–304. doi: 10.1093/eurjpc/zwac263
 57. Kawachi S, Takeda N, Sasaki A, Kokubo Y, Takami K, Sarui H, Hayashi M, Yamakita N, Yasuda K. Circulating insulin-like growth factor-1 and insulin-like growth factor binding protein-3 are associated with early carotid atherosclerosis. *Arterioscler Thromb Vasc Biol*. 2005;25:617–621. doi: 10.1161/01.ATV.00000154486.03017.35
 58. Backman JD, Li AH, Marcketta A, Sun D, Mbatchou J, Kessler MD, Benner C, Liu D, Locke AE, Balasubramanian S, et al; Regeneron Genetics Center. Exome sequencing and analysis of 454,787 UK Biobank participants. *Nature*. 2021;599:628–634. doi: 10.1038/s41586-021-04103-z
 59. Giri A, Hellwege JN, Keaton JM, Park J, Qiu C, Warren HR, Torstenson ES, Kovesdy CP, Sun YV, Wilson OD, et al; Understanding Society Scientific Group. Trans-ethnic association study of blood pressure determinants in over 750,000 individuals. *Nat Genet*. 2019;51:51–62. doi: 10.1038/s41588-018-0303-9
 60. He P, Zhang H, Yun CC. IRBIT, inositol 1,4,5-triphosphate (IP3) receptor-binding protein released with IP3, binds Na⁺/H⁺ exchanger NHE3 and activates NHE3 activity in response to calcium. *J Biol Chem*. 2008;283:33544–33553. doi: 10.1074/jbc.M805534200
 61. Kobayashi K, Monkawa T, Hayashi M, Saruta T. Expression of the Na⁺/H⁺ exchanger regulatory protein family in genetically hypertensive rats. *J Hypertens*. 2004;22:1723–1730. doi: 10.1097/00004872-200409000-00016
 62. Kim YT, Huh JW, Choi YH, Yoon HK, Nguyen TT, Chun E, Jeong G, Park S, Ahn S, Lee WK, et al. Highly secreted tryptophanyl tRNA synthetase 1 as a potential theranostic target for hypercytokinemic severe sepsis. *EMBO Mol Med*. 2023;16:40–63. doi: 10.1038/s44321-023-00004-y
 63. Tokolyi A, Persyn E, Nath AP, Burnham KL, Marten J, Vanderstichele T, Tardaguila M, Stacey D, Farr B, Iyer V, et al. The contribution of genetic determinants of blood gene expression and splicing to molecular phenotypes and health outcomes. *Nat Genet*. 2025;57:616–625. doi: 10.1038/s41588-025-02096-3
 64. Herdenberg C, Mutie PM, Billing O, Abdullah A, Strawbridge RJ, Dahlman I, Tuck S, Holmlund C, Arner P, Henriksson R, et al. LRIG proteins regulate lipid metabolism via BMP signaling and affect the risk of type 2 diabetes. *Commun Biol*. 2021;4:90. doi: 10.1038/s42003-020-01613-w
 65. Moon J-S, Ho C-C, Park J-H, Park K, Shin B-Y, Lee S-H, Sequeira I, Mun CH, Shin J-S, Kim J-H, et al. Lig1-expression confers suppressive function to CD4⁺ cells and is essential for averting autoimmunity via the Smad2/3/Foxp3 axis. *Nat Commun*. 2023;14:5382. doi: 10.1038/s41467-023-40986-4
 66. Hirohama D, Fadista J, Ha E, Liu H, Abedini A, Levinsohn J, Vassalotti A, Zeng L, Li C, Mohandes S, et al. The proteogenomic landscape of the human kidney and implications for cardio-kidney-metabolic health. *Nat Med*. 2025;31:3917–3929. doi: 10.1038/s41591-025-03872-8
 67. Simonds SE, Pryor JT, Ravussin E, Greenway FL, Dileone R, Allen AM, Bassi J, Elmquist JK, Keogh JM, Henning E, et al. Leptin mediates the increase in blood pressure associated with obesity. *Cell*. 2014;159:1404–1416. doi: 10.1016/j.cell.2014.10.058
 68. Arnold N, Hermanns IM, Schulz A, Hahad O, Schmitt VH, Panova-Noeva M, Prochaska JH, Binder H, Pfeiffer N, Beutel M, et al. Renin, aldosterone, the aldosterone-to-renin ratio, and incident hypertension among normotensive subjects from the general population. *Cardiovasc Res*. 2022;119:294–301. doi: 10.1093/cvr/cvac019
 69. Sahay M, Sahay RK. Low renin hypertension. *Indian J Endocrinol Metab*. 2012;16:728–739. doi: 10.4103/2230-8210.100665
 70. Preissner KT, Reuning U. Vitronectin in vascular context: facets of a multitasked matricellular protein. *Semin Thromb Hemost*. 2011;37:408–424. doi: 10.1055/s-0031-1276590
 71. Malinova TS, Angulo-Urarte A, Nüchel J, Tauber M, van der Stoep MM, Janssen V, de Haan A, Groenen AG, Tebbens M, Graupera M, et al. A junctional PACSIN2/EHD4/MICAL-L1 complex coordinates VE-cadherin trafficking for endothelial migration and angiogenesis. *Nat Commun*. 2021;12:2610. doi: 10.1038/s41467-021-22873-y
 72. Schoeler T, Speed D, Porcu E, Pirastu N, Pingault J-B, Kutalik Z. Participation bias in the UK Biobank distorts genetic associations and downstream analyses. *Nat Hum Behav*. 2023;7:1216–1227. doi: 10.1038/s41562-023-01579-9